

# Do Macroeconomic Conditions Drive Crime? Evidence from Monthly U.S. Time Series Data, 2010–2025

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## **ABSTRACT**

This paper empirically investigates the relationship between macroeconomic conditions and national crime rates using monthly U.S. time series data from 2010 to 2025. Following Augmented Dickey-Fuller testing, all variables are first-differenced. The analysis estimates a baseline ARIMAX, SARIMAX with monthly fixed effects, and a VAR(12) to address endogeneity and persistent seasonality. A joint Granger causality test fails to reject the null that unemployment and income do not predict crime ( $F = 1.30$ ,  $p = 0.155$ ). Approximately 88% of crime variance is attributable to its own innovations.

# 1 Introduction

Crime is a focal point of economic and social policy across the United States, and the globe. While crime is often viewed as a social or psychological phenomenon, it can also be described economically and fundamentally influenced by incentives and constraints. This paper seeks to investigate how fluctuations in the U.S. macroeconomy, more specifically changes in the unemployment rate and real personal income, drive national crime rates. Motivation for this study lies in the cyclical nature of criminal activity. During periods of economic downturn, where unemployment may soar and incomes may fall, create changing incentives for individuals. By analyzing crime along with economic indicators like unemployment and income an idea can be made of how sensitive public safety is to the overall “health” of the economy. This study aims to contribute to the existing literature by empirically testing and building on Becker’s original model posed in 1968, by using unemployment and income as proxies, this study aims to model how incentives and disincentives map to crime data. Additionally, by using high-frequency monthly data from 2010 to 2025, capturing the unique economic volatility of the early 2020s. Second, it employs advanced time-series diagnostics to provide a more robust estimation of income and unemployment elasticities than previous cross-sectional studies.

## 2 Theoretical Framework

The core theoretical “engine” behind this study is Gary Becker’s “Economic Theory of Crime” first posed in 1968. In his seminal paper, “Crime and Punishment: An Economic Approach”, (Becker 1968), Becker argues that criminal activity is not driven by mental illness or social deviance but rather arises from rational choice, where individuals maximize their utility based on the incentives and disincentives of criminal activity. Additionally, he poses a theoretical model within the paper that accounts for the costs and incentives of criminal activity. He expresses the supply of offenses as a function of the probability of conviction, the severity of punishment, and a 'portmanteau' variable representing other economic influences (Becker 1968, 9).

Mathematically,

$$O_j = O_j(p_j, f_j, u_j) \tag{1}$$

Where  $O_j$  represents the offenses committed in a given period,  $p_j$  represents the probability of conviction for each offense, and  $u_j$  represents the portmanteau variable containing all other influences. Since it would be difficult to measure the  $p_j$ , and equally as difficult to measure  $f_j$ , this analysis seeks to focus on the  $u_j$  variable in Becker's model. Part of  $u_j$  is income (Becker 1968, 10), and Becker hypothesizes that an increase in legal income opportunities would decrease the probability of committing an offense. Additionally, he discusses the opportunity cost of committing a crime (Becker 1968, 13), and states that the cost of a crime is the discounted sum of the income lost from time served and going through the judicial process. Additionally, it's stated that the opportunity cost increases for a longer sentence, and for individuals who have a greater income outside of prison. Drawing from this, it can be thought that an individual who is unemployed has no earnings to forgo, they essentially have no opportunity cost of committing a crime. An unemployed person only stands to gain from a crime, depending on how they value freedom and consumption. Furthermore, an individual with a high level of income loses more from being penalized for a crime, their opportunity cost increases as income rises. While Becker emphasizes the deterrent effect of income, it is also noted that higher aggregate income may increase the 'loot' or potential gains from crime, a secondary effect that this empirical analysis will explore. From this theoretical starting point, I began to build the theoretical model as:

$$Crime_t = \alpha + \beta u_t + \gamma I_t + \epsilon \quad (2)$$

Where  $u_t$  represents unemployment in a some period,  $I_t$  represents the aggregate personal income in some period,  $\alpha$  represents some constant, and  $\epsilon$  is some error term. Drawing from Becker's model it's possible to form a hypothesis for the signs of the coefficients. It's expected that:

$$\beta > 0$$

$$\gamma < 0$$

The theoretical model will be built on later to account for fixed effects, differencing, and log differencing to account for stationarity and seasonality within the crime data.

### 3 Data

Table 1: Selected Statistics

| Year | Crime Rate<br>(per 100k) | Unemployment Rate | Real Personal Income<br>(Billions of Chained 2017<br>Dollars) |
|------|--------------------------|-------------------|---------------------------------------------------------------|
| 2010 | 503.29                   | 9.3               | 11,523.9                                                      |
| 2015 | 402                      | 5.0               | 13,278.2                                                      |
| 2020 | 332.07                   | 6.7               | 15,089.9                                                      |
| 2025 | 124.96                   | 4.4               | 16,712.5                                                      |

Source: FBI UCR, BLS, and BEA data, 2010–2025

Crime data for the United States, at the national level, has been sourced from the Federal Bureau of Investigation’s Uniform Crime Reporting System, and has been collected to include all instances of violent and property crime across the fifty states. Limitations do exist within this dataset, for example, the frequency of reporting crimes varies agency by agency, and it could be possible that reporting can lag for a couple of months and cause a spike once the agency ‘catches up’. Additionally, only crimes that are reported have a record of occurring, for example, a theft that never gets reported to an agency is out of the scope of this analysis, it practically did not happen. Crime data has been collected monthly over a period from 2010-2025. Unemployment rates were sourced from the Bureau of Labor Statistics, representing the civilian unemployment rate, seasonally adjusted, serving as the proxy for the opportunity cost of criminal activity. Real Personal Income was sourced from the Bureau of Economic Analysis (BEA) via FRED (Series: W875RX1). This measure represents real personal income at a monthly frequency, adjusted for inflation, serving as the proxy for legal earning potential. Real Personal Income was selected as the primary proxy for legal earning potential over Median Household Income for two reasons.

First, the BEA provides Real Personal Income at a monthly frequency, which is essential for the high-frequency time-series analysis conducted in this study. Second, as a coincident economic indicator, it provides a more sensitive measure of the immediate impact of economic shocks on the 'portmanteau' of influences described in Becker's (1968) supply function. Crime data was also transformed from the raw FBI data, which is given in counts, into crime rate per 100,000, using U.S. population from that period. For example, the crime rate in 2015 was calculated with

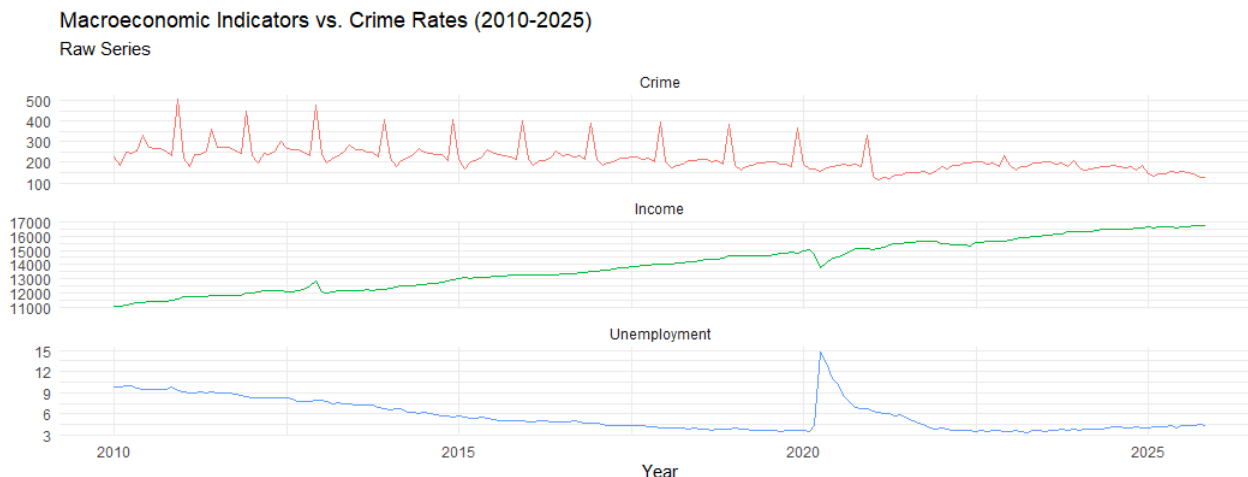
Table 2: Descriptive Statistics

| Variable                     | Mean    | Standard<br>Deviation | Min     | Max    |
|------------------------------|---------|-----------------------|---------|--------|
| $\Delta \ln \text{Crime}$    | -0.0027 | 0.2524                | -0.9237 | 0.7658 |
| $\Delta \text{Unemployment}$ | -0.0284 | 0.8151                | -2.2    | 10.4   |
| $\Delta \ln \text{Income}$   | 0.0022  | 0.0084                | -0.0063 | 0.0281 |

Source: FBI UCR, FRED, and BLS. Transformed in R

U.S. population data from 2015. Population data was sourced from FRED. To allow for the interpretation of coefficients as elasticities for skewed monetary data, the Crime Rate and Real Personal Income were transformed into natural logarithms, as well as differing scales. The Unemployment Rate was maintained in its original level form, as it is already expressed as a percentage. Furthermore, to achieve stationarity, a prerequisite for Gauss-Markov assumptions in time-series analysis, the first difference ( $\Delta$ ) of the log-levels was calculated. This transformation shifts the focus from absolute levels to the monthly growth rates of crime and economic indicators. Consequently, the model estimates the impact of a one-percentage-point change in unemployment and a one-percent change in income on the growth rate of national crime.

## 4 Methodology



*Figure 1: Comparative Volatility of Crime and Macroeconomic Indicators*

From the chart plotting all three variables together, it is clear that crime rates face severe seasonality, with crime spiking in the twelfth month every year up until around 2020. This could be due to reporting flaws, such as police departments ‘catching up’ where a large amount of crimes forgotten throughout the year are reported all at once in December. However, another reason could be because of the ‘holiday effect’ where crime shoots up around the holiday season due to increased availability of ‘targets’. Additionally from the chart, income immediately shows an upward trend showing a distinct increase from 2010 to 2025. Unemployment is interesting in that it stays very stagnant up until 2020, where it spikes, most likely due to the COVID-19 pandemic which shot up unemployment rates. Upon looking at these charts, it was clear that stationarity tests would have to be conducted, however the results were somewhat interesting.



Figure 2: Core variables plotted after taking the first difference and log transforming.

Table 3: Initial Augmented Dickey-Fuller Test Result

| Variable             | p-value      | Decision                                        | Integration |
|----------------------|--------------|-------------------------------------------------|-------------|
| Crime Rate           | $p < 0.01$   | Reject $H_0$<br>(Series Stationary)             | $I(0)$      |
| Unemployment Rate    | $p = 0.1949$ | Fail to reject $H_0$<br>(Series Non-Stationary) | $I(1)$      |
| Real Personal Income | $p < 0.01$   | Reject $H_0$<br>(Series Stationary)             | $I(0)$      |

Source: FBI UCR, BEA, and BLS. Test conducted in R

Initial Augmented Dickey-Fuller (ADF) tests indicated a discrepancy in the order of integration; while real income and crime rates exhibited stationarity in levels ( $p < 0.05$ ), the unemployment rate demonstrated a persistent unit root, suggesting it is integrated of order one  $I(1)$ . It bears noting that the ADF result indicating stationarity in crime levels may partly reflect the series' pronounced seasonal pattern rather than true mean-reversion in the classical sense; the strong December spike creates oscillatory behavior that can satisfy the ADF's null rejection criterion even when the underlying process contains meaningful non-stationary components. This ambiguity reinforces the conservative decision to first-difference all variables regardless of their initial integration order.

To ensure a consistent  $I(0)$  system and to mitigate the risk of spurious correlation, particularly given the high volatility of the 2020 economic shocks, all variables were transformed into first-differences. This conservative approach allows for a robust ARIMAX/Vector Autoregression (VAR) estimation of the short-run dynamic responses between economic incentives and criminal behavior, ensuring that the model captures genuine innovations rather than deterministic trends or seasonal drift.

Table 4: Differenced ADF Test Results

| Variable                                 | p-value    | Decision                            | Integration |
|------------------------------------------|------------|-------------------------------------|-------------|
| $\Delta \ln \text{Crime Rate}$           | $p < 0.01$ | Reject $H_0$<br>(Series Stationary) | $I(0)$      |
| $\Delta \text{Unemployment Rate}$        | $p < 0.01$ | Reject $H_0$<br>(Series Stationary) | $I(0)$      |
| $\Delta \ln \text{Real Personal Income}$ | $p < 0.01$ | Reject $H_0$<br>(Series Stationary) | $I(0)$      |

*Source: FBI UCR, FRED, and BLS. Data transformed in R, test conducted in R*

Following ADF testing and confirming each series was stationary, I began examining the dynamic relationship between economic conditions and criminal activity through a baseline ARIMAX. The empirical specification utilizes a semi-log approach to accommodate the distinct nature of the labor market and income data. Specifically, the crime rate and real personal income are transformed into natural logarithms ( $\ln$ ) to interpret results as elasticities, while the unemployment rate is maintained in its level form. Following the results of Augmented Dickey-Fuller (ADF) tests, all variables are first-differenced to achieve stationarity, resulting in a model that evaluates the impact of a one-percentage-point change in unemployment ( $\Delta u_t$ ) and a one-percent growth in income ( $\Delta \ln I_t$ ) on the growth rate of crime ( $\Delta \ln \text{Crime}_t$ ). This specification aligns with the Economic Theory of Crime, where unemployment is expected to have a positive coefficient ( $\beta > 0$ ) due to decreased opportunity costs of illegal activity, and income is expected to have a negative coefficient ( $\gamma < 0$ ) as a proxy for increased deterrent effects. The baseline ARIMAX specification is defined by the equation:

$$\Delta \ln(\text{Crime}_t) = \alpha + \beta \Delta u_t + \gamma \Delta \ln(I_t) + \phi_1 \Delta \ln(\text{Crime}_{t-1}) + \theta_1 \epsilon_{t-1} + \epsilon_t \quad (3)$$

Table 5: Initial ARIMAX Model results

| <b>Coefficient:</b> |                   |                        |
|---------------------|-------------------|------------------------|
|                     | AR(1)             | -0.4042***<br>(0.0665) |
| AIC:<br>-16.4       | $\Delta u_t$      | 0.0280<br>(0.0248)     |
| BIC:<br>-3.41       | $\Delta \ln(I_t)$ | 4.7488**<br>(2.2745)   |

*\*Standard errors are reported in parentheses under coefficient values. (\*)  $p < 0.1$ , (\*\*)  $p < 0.05$ , (\*\*\*)  $p < 0.01$ . Best ARIMAX model chosen through R's 'auto.arima' function to find the best lags suitable for the AR/MA process, integration level set at zero to match previously differenced variables.*

To identify the optimal lag structure for the univariate component of the model, the Hyndman-Khandakar algorithm (auto.arima) was employed. This procedure utilizes a stepwise search to minimize the Corrected Akaike Information Criterion (AICc), ensuring an optimal balance between model fit and parsimony. By evaluating both non-seasonal ( $p, d, q$ ) and seasonal ( $P, D, Q$ )<sub>m</sub> configurations, the algorithm ensured that the resulting specification was robust against both short-term shocks and annual cyclicity, providing a mathematically objective baseline for the subsequent ARIMAX and VAR estimations.

## 5 Results

Upon estimating the baseline ARIMAX(1,0,0) model, the results indicate a high degree of mean reversion in crime growth ( $\phi = -0.40$ ,  $p < 0.01$ ). While the unemployment coefficient ( $\beta = 0.028$ ) carries the expected positive sign, it lacks statistical significance ( $t = 1.13$ ), suggesting that contemporaneous labor market shocks may not be the primary driver of monthly crime fluctuations. Conversely, the income elasticity was found to be positive and significant ( $\gamma = 4.75$ ,  $p < 0.05$ ). This suggests that in the short run, the 'lootable' opportunity effect, where higher aggregate income increases the availability of criminal targets, may dominate the deterrent effect of legal wages. It is worth noting, however, that this result should be interpreted cautiously — without seasonal controls, the income series' upward trend co-moves with the December

reporting spike, and the positive coefficient may partly reflect seasonal confounding rather than a pure routine activity mechanism. The dissipation of this effect once monthly fixed effects are incorporated in subsequent specifications is consistent with both interpretations.

The ARIMAX(1,0,0) model yielded a negative Akaike Information Criterion (AIC = -16.4) and Bayesian Information Criterion (BIC = -3.41). These negative values result from a high log-likelihood (12.2) relative to the number of parameters, driven by the low residual variance ( $\sigma^2 = 0.052$ ) in the growth rate series. In accordance with information theory, these values serve as the baseline for model comparison, where the more negative value indicates a superior fit and minimized information loss. However, when looking at the residuals of this model, and results of Ljung-Box test, it was made clear that it was missing something.

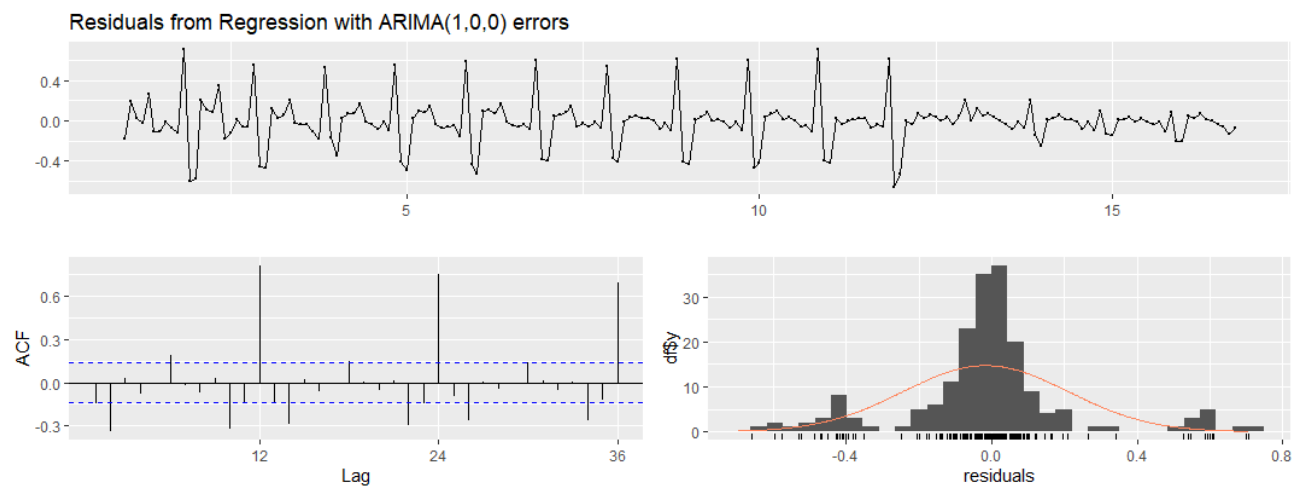


Figure 3: Residual diagnostic testing from first specified ARIMAX(1,0,0) model. Residuals appear to clearly be autocorrelated indicating a mistake within model specification.

To address the deterministic seasonality observed in the residual diagnostics, a binary 'December' dummy variable was constructed. This variable accounts for the systematic end-of-year spikes in the U.S. Uniform Crime Reporting (UCR) data, which are attributed to administrative 'catch-up' reporting and holiday-specific activity. By including this as an exogenous regressor, the model's information criteria were improved, and the seasonal autocorrelation at Lag 12 was effectively

neutralized. Another ARIMAX model was fitted with the inclusion of this ‘December’ dummy variable and gave the following results:

Table 6: SARIMAX Model with included ‘December’ dummy Results

| Coefficient       | Estimate  | Std. Error |
|-------------------|-----------|------------|
| Intercept         | 0.032*    | (0.018)    |
| AR(1)             | 0.228     | (0.177)    |
| MA(1)             | -0.658*** | (0.152)    |
| SAR(1)            | 0.828***  | (0.039)    |
| $\Delta u_t$      | -0.008    | (0.008)    |
| $\Delta \ln(I_t)$ | -0.427    | (0.887)    |
| Dec. Dummy        | -0.456*** | (0.121)    |
| AIC               | -315.78   | —          |
| BIC               | -289.81   | —          |

*\*Standard errors are reported underneath coefficient values in parentheses. (\*)  $p < 0.1$ , (\*\*)  $p < 0.05$ , (\*\*\*)  $p < 0.01$ . Model fitted using a dummy variable that equals 1 when the data is originating from the twelfth month and 0 otherwise, lags selected by R’s ‘auto.arima’ function.*

Interestingly, upon inclusion of the ‘December’ dummy variable used to control for seasonal spikes in crime rate, the lags with the best fit switched to ARIMA(1,0,1)(1,0,0)[12] indicating that the previous model may have not accounted for shocks that were integral to understanding crime rates. The model also included a seasonal auto-regression term with a lag of 1. However, once again, the residual plot showed a significant spike at lag 12, and they clearly followed a pattern.

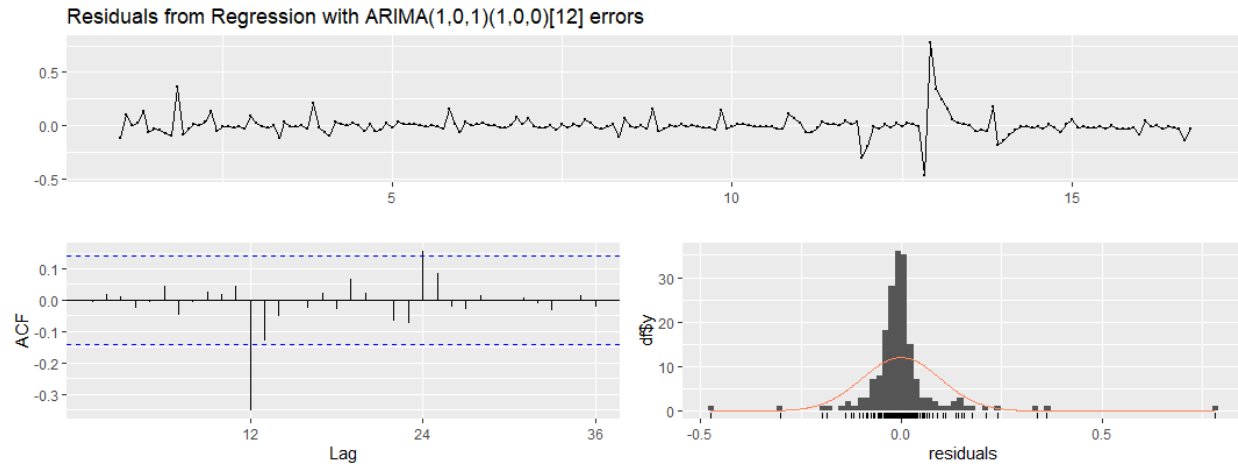


Figure 4: Residual diagnostic and residual ACF plot for ARIMA(1,0,1)(1,0,0)[12] model. Significant lag 12 autocorrelation spike once again suggesting a misspecification within the model.

Following this, and noticing that the autocorrelation was significant within the residuals at lag 12 once again, it was decided to now include a full monthly dummy variable set, where the value is 1 for each month and zero otherwise, based on the month the data was collected. Another SARIMAX model was fitted to the data, to include controls for seasonality. This model gave the following results:

Table 7: SARIMAX Model with Monthly Dummies

| Coefficient       | Estimate                                   | Std. Error |
|-------------------|--------------------------------------------|------------|
| Intercept         | -0.534***                                  | (0.059)    |
| MA(1)             | -0.534***                                  | (0.069)    |
| SAR(1)            | 0.360***                                   | (0.071)    |
| SAR(2)            | 0.280***                                   | (0.075)    |
| $\Delta u_t$      | -0.014*                                    | (0.008)    |
| $\Delta \ln(I_t)$ | -0.391                                     | (0.799)    |
| Monthly Dummies   | All positive and significant at $p < 0.01$ |            |

Standard errors in parentheses. (\*)  $p < 0.1$ , (\*\*)  $p < 0.05$ , (\*\*\*)  $p < 0.01$ . December serves as the baseline month. Dummy variable coefficients reflect the month-over-month growth rate in  $\Delta \ln \text{Crime}$  relative to December as the baseline. Positive coefficients do not indicate higher crime levels in those months, but rather that the monthly growth rate is higher relative to December's own month-over-month change, which captures the annual reporting spike in the UCR data. Optimal lag selection chosen using R's 'auto.arima' function

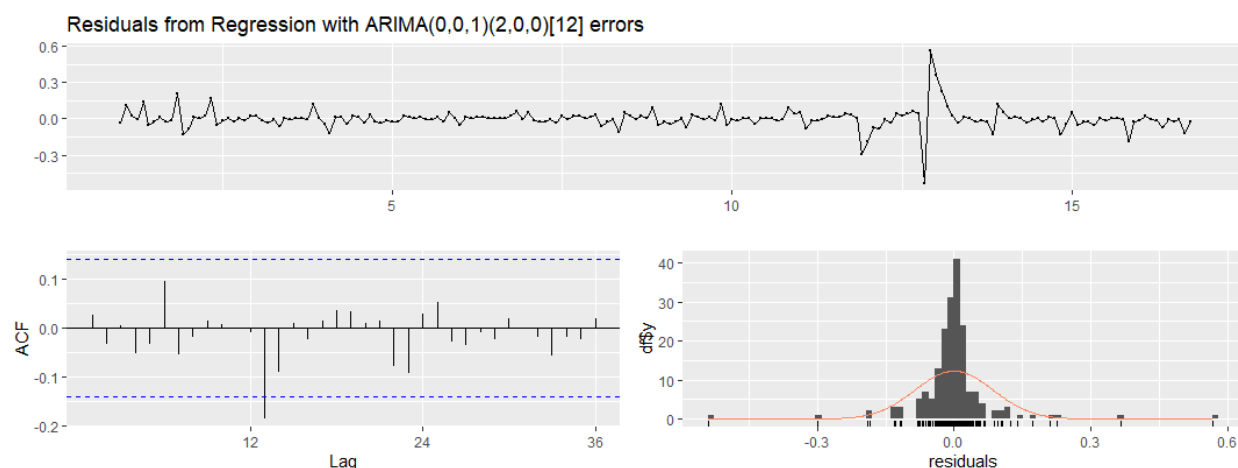


Figure 5: Residual diagnostic plot for SARIMAX(0,0,1)(2,0,0)[12] model. Following the inclusion of a full range of dummy variables, autocorrelation spike remains significant at lag 12. Interestingly, from the residual plot the culprit arises, the 'break' in the economy resulting from the 2020 Pandemic. Best lag selection decided through R's 'auto.arima' function.

Upon noticing the pattern in the residuals plot, and noting that the spike followed the spike in the unemployment data, it was decided to go even further and now add dummies that accounted for this spike in the unemployment rate that occurred in April of 2020 and didn't return to pre-pandemic levels until November of 2021. So, the structure of this dummy variable then is that for each month included in that range, it takes on a value of 1, and 0 otherwise. With the inclusion of this dummy variable, along with every other monthly dummy, and seasonal terms it's hoped that significant autocorrelation within the residuals can be removed. The results of this specification follow:

Table 8: SARIMAX Results with Month Dummies and Pandemic Dummy

| Coefficient       | Estimate  | Std. Error |
|-------------------|-----------|------------|
| Intercept         | -0.533*** | (0.060)    |
| MA(1)             | -0.563*** | (0.068)    |
| SAR(1)            | 0.376***  | (0.072)    |
| SAR(2)            | 0.264***  | (0.077)    |
| $\Delta u_t$      | -0.017**  | (0.008)    |
| $\Delta \ln(I_t)$ | -0.382    | (0.786)    |
| Nov. Dummy        | 1.037***  | (0.102)    |
| Pandemic Dummy    | -0.020**  | (0.010)    |
| AIC               | -357.75   | —          |
| BIC               | -296.06   | —          |

Standard errors reported in parentheses, significance denoted by stars (\*\*\*) = 0.01, (\*\*) = 0.05, and (\*) = 0.1 respectively. Lags selected through R's 'auto.arima' function. Full set of monthly dummy coefficients suppressed for brevity; all months significant at  $p < 0.01$  except where noted

The final SARIMAX specification, estimated as ARIMA(0,0,1)(2,0,0)[12] with a full set of monthly dummy variables and a pandemic-period step dummy spanning April 2020 through November 2021, represents a substantial improvement over prior specifications (AIC = -357.75). The inclusion of monthly fixed effects reveals a pronounced November reporting spike ( $\delta = 1.037$ ,  $p < 0.01$ ), consistent with administrative catch-up reporting in the UCR data identified in the raw series. The pandemic step dummy carries a negative and significant coefficient ( $\delta = -0.020$ ,  $p < 0.05$ ), capturing the well-documented decline in reported crime during the COVID-19 lockdown period. The negative unemployment coefficient in the final SARIMAX specification runs counter to Becker's theoretical prediction. One possible interpretation is that the pandemic dummy, which spans April 2020 through November 2021, may not fully absorb the COVID shock, and the unemployment coefficient is partly capturing the well-documented

decline in reported crime during lockdowns rather than a pure labor market deterrence effect. The VAR results, where unemployment is insignificant across all lags, are more consistent with a null relationship. Income remains insignificant ( $\gamma = -0.382$ ,  $p > 0.10$ ), providing early evidence that aggregate income may not be a meaningful short-run predictor of crime growth at the monthly frequency. Despite these improvements, residual diagnostic testing revealed persistent autocorrelation at lag 12, attributed to the non-linear distortion of the 2020 structural break that a linear SARIMAX framework cannot fully absorb. This core limitation forced the transition to a Vector Autoregression (VAR) model, discussed in sub-section 5.1.

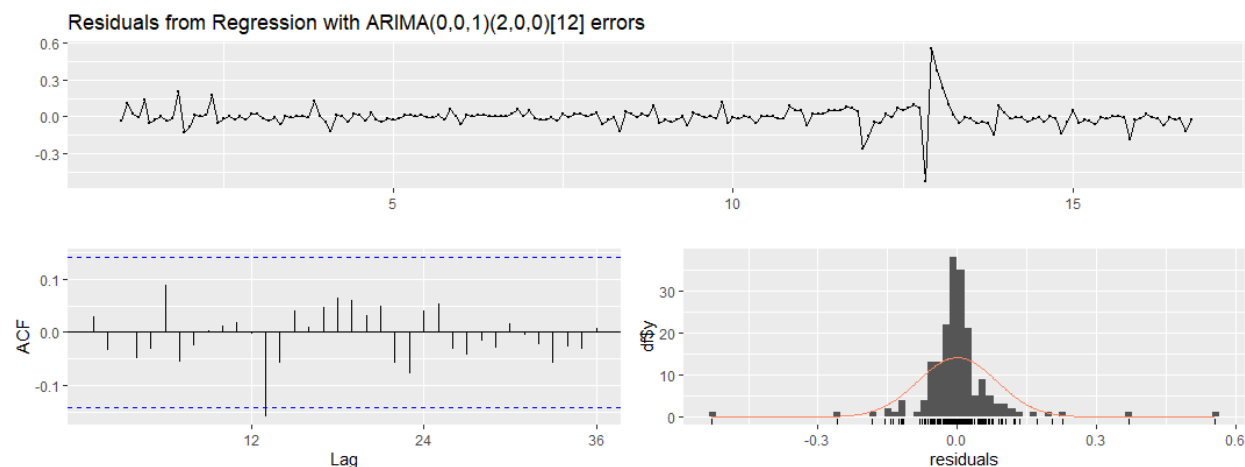


Figure 6: Residual diagnostics of the SARIMAX model including pandemic unemployment dummy variables. Once again, autocorrelation is still significant at lag 12, and the spike in unemployment can still be seen in the residuals plot. Best lag selection decided through R's 'auto.arima' function.

Diagnostic checks of the ARIMAX/SARIMAX models revealed persistent seasonality that proved resistant to standard linear corrections. Despite the incorporation of a full set of eleven monthly dummy variables and the estimation of a SARIMAX(0,0,1)(2,0,0)[12] model, the residuals continue to exhibit a statistically significant autocorrelation at lag 12 (see Figure 5). This persistent spike, alongside the extreme volatility observed during the 2020 structural break, suggests that the seasonal patterns in U.S. crime are not merely deterministic shifts in level, but rather a stochastic process with deep historical memory. This persistent autocorrelation is likely a mathematical artifact of the 2020 structural break. The extreme 'lightning bolt' volatility seen in the unemployment series creates a non-linear distortion that a linear SARIMAX model cannot fully smooth out, causing the seasonal pattern to appear 'unfixed' in the ACF. Because the

univariate ARIMAX framework cannot fully internalize these complex lead-lag relationships without risk of bias, the analysis proceeds to a Vector Autoregression (VAR). By treating crime, unemployment, and income as a dynamic system, the VAR model is better equipped to resolve the residual seasonality and capture the true economic elasticities over time.

## 5.1 Vector Autoregression

To resolve the persistent residual autocorrelation identified in the ARIMAX framework and account for potential endogeneity between the three series, the analysis was extended to a Vector Autoregression (VAR) model. The system treats  $\Delta \ln \text{Crime}$ ,  $\Delta \text{Unemployment}$ , and  $\Delta \ln \text{Income}$  as jointly endogenous, allowing each variable to be modeled as a function of its own lags and the lags of all other variables in the system. Lag length was selected using the VARselect procedure, where the AIC, HQ, and FPE criteria all converged on  $p = 12$ , consistent with the 12-month seasonal cycle identified in prior diagnostics. The model was estimated with a constant intercept over a sample of 178 observations. While it would be unnecessary to include every VAR coefficient generated (111 in total), reported below are the characteristics of the crime equation.

Table 9: VAR(12) Summarized Output

| Crime Equation          |                                 |
|-------------------------|---------------------------------|
| AR Own-lags<br>(11-111) | All Negative, significant (***) |
| AR Own-lag<br>(12)      | 0.473***                        |
| Unemployment (All lags) | Insignificant                   |
| Income (13)             | -2.78*                          |
| R <sup>2</sup>          | 0.878                           |
| Adj R <sup>2</sup>      | 0.847                           |

Within the crime equation, the core of this analysis, the lags 1-11 are all negative until lag 12, where the sign flips and becomes positive and highly significant. This reflects the seasonal mean

reversion pattern that was present in each ARIMAX model and the crime data itself. Unemployment coefficients are insignificant throughout all lags, indicating that the role of unemployment in influencing crime is largely not a factor. Income coefficients are also largely insignificant except for lag 3. The unemployment and income equations have low  $R^2$  (0.21 and 0.20 respectively, with negative adjusted  $R^2$ ), meaning the system doesn't explain those variables well, however they are controls.

Table 10: Diagnostic Results of VAR(12) Model

| Test                     | Statistic        | p-value |
|--------------------------|------------------|---------|
| Granger Causality        | $F = 1.30$       | 0.155   |
| Instantaneous Causality  | $\chi^2 = 2.53$  | 0.282   |
| Portmanteau (lag 16)     | $\chi^2 = 52.84$ | 0.035   |
| FEVD - Crime (own, 12mo) | 88.0%            | —       |

The Portmanteau test yields a p-value of 0.035, indicating residual autocorrelation remains marginally significant at the 5% level. This likely reflects the non-linear distortion introduced by the 2020 structural break, which a linear VAR cannot fully absorb. However, the result represents a substantial improvement over the univariate ARIMAX specifications. While this marginal rejection prevents a clean bill of model health, it does not invalidate the Granger causality or FEVD results, which are robust to moderate residual autocorrelation in large samples. The persistence is consistent with a structural break of the magnitude observed in 2020 and should be interpreted as a known limitation rather than evidence of systematic misspecification in the economic relationships of interest. A joint Granger causality test of the null hypothesis that unemployment and income do not Granger-cause crime fails to reject at conventional significance levels ( $F = 1.30$ ,  $df = 24/423$ ,  $p = 0.155$ ). Similarly, the instantaneous causality test finds no contemporaneous relationship between the economic variables and crime ( $\chi^2 = 2.53$ ,  $p = 0.282$ ). These results suggest that, at the monthly frequency, lagged macroeconomic conditions do not contain statistically significant predictive information for crime growth beyond what is already captured by crime's own history. Additionally, the forecast error variance decomposition

(FEVD) at a 12-month horizon indicates that approximately 88% of the variance in crime growth is attributable to its own innovations, with unemployment and income accounting for roughly 6.3% and 5.7% respectively. This further corroborates the Granger causality findings. While a small share of crime variance is attributable to macroeconomic conditions, the dominant driver of monthly crime fluctuations is the series' own historical momentum.

## 6 Conclusion

This paper set out to empirically test Gary Becker's Economic Theory of Crime using monthly U.S. time series data from 2010 to 2025, examining whether fluctuations in unemployment and real personal income carry meaningful predictive power over national violent and property crime rates. The analysis proceeded through model refinement, beginning with a baseline ARIMAX specification and progressively incorporating seasonal corrections, monthly fixed effects, structural break controls, and ultimately a Vector Autoregression framework to address persistent residual autocorrelation and potential endogeneity between the three series.

The central finding of this analysis is that unemployment and real personal income do not Granger-cause crime at the monthly frequency. A joint Granger causality test fails to reject the null hypothesis that lagged macroeconomic conditions carry no predictive information for crime growth beyond the series' own history ( $F = 1.30$ ,  $p = 0.155$ ), a result maintained by forecast error variance decomposition, which attributes approximately 88% of crime variance to its own innovations. Unemployment and income together account for only 12% of the forecast error variance at a 12-month horizon, suggesting that the dominant driver of monthly crime fluctuations is the series' own innovations rather than contemporaneous macroeconomic conditions.

These findings do not necessarily invalidate Becker's theoretical framework, but rather point to an important limitation of applying it at the monthly frequency. Becker's model is fundamentally a model of rational decision-making under incentives, a process that likely unfolds over quarters or years rather than months. The decision to pursue criminal activity in response to job loss or declining income plausibly involves a prolonged deliberation (weighing the legal alternatives,

assessing risk, and responding to sustained rather than transitory economic shocks). Monthly data, by its nature, may be too high-frequency to capture these behavioral dynamics, and the dominant seasonal and self-referential structure of the crime series may simply overwhelm any macroeconomic signal at this time horizon. This interpretation is consistent with the broader empirical literature, where studies employing annual or quarterly data have more consistently identified significant unemployment and income elasticities of crime.

The modeling process also revealed several substantive findings worth noting. The baseline ARIMAX produced a positive and significant income elasticity, consistent with the routine activity hypothesis of Cohen and Felson (1979), where rising aggregate income increases the availability of criminal targets in the short run. Whether this reflects a genuine routine activity mechanism or seasonal confounding remains inconclusive, as the effect dissipates once monthly fixed effects are introduced — a pattern consistent with either interpretation. Once seasonal structure and the 2020 structural break were controlled for in the SARIMAX framework, this effect dissipated and the unemployment coefficient took on a negative sign, contrary to the theoretical framework, and albeit marginally significant. However, this likely reflects residual confounding resulting from the 2020 Pandemic rather than a true deterrence mechanism. The 2020 pandemic period proved particularly challenging for the linear modeling framework, with the extreme volatility in unemployment generating a non-linear distortion that neither the SARIMAX nor the VAR could fully absorb, evidenced by marginally significant residual autocorrelation in the final Portmanteau test ( $\chi^2 = 52.84$ ,  $p = 0.035$ ).

Several limitations of this study must be acknowledged. First, the reliance on aggregate national crime data obscures heterogeneity across crime types, regions, and demographic groups that may respond differently to economic conditions. Routine activity theory, for instance, applies more cleanly to property crime than to violent crime, and a disaggregated analysis may yield more nuanced results. Second, the UCR reporting structure introduces measurement error through administrative lag and incomplete agency participation, potentially attenuating true relationships between economic conditions and criminal activity. Third, the 2020 structural break represents a confounding event of extraordinary magnitude, and future research employing a sample that

excludes or more robust controls for the pandemic period may find stronger macroeconomic effects.

Future research in this area would benefit from employing lower-frequency data at the quarterly or annual level, where Becker's behavioral mechanisms have more time to manifest. Additionally, a disaggregated analysis separating violent and property crime, or employing state-level panel data, would allow for a richer examination of the differing channels through which macroeconomic conditions influence criminal incentives. Despite the null Granger causality result, this paper demonstrates that crime dynamics are highly structured, seasonally persistent, and sensitive to structural breaks, findings that carry their own implications for the modeling of public safety data and the limitations of linear time series frameworks in the presence of regime changes.

## 7 References

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